

KRANGEDE, Sweden

WMO: 022470

Lat: 63.1512N

Lon: 16.1699E

Elev: 604

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-10.4	-21.3	1.7	-15.5	-15.5	2.4	-10.3	15.9	31.7	13.4	31.5	0.6	330	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	78.5	62.4	74.9	61.1	71.4	59.5	65.0	74.2	62.9	71.4	61.0	68.7	6.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	83.7	68.1	59.4	77.5	65.8	57.6	72.5	64.3	30.3	74.1	28.7	71.7	27.4	68.8	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.2	10.5	DB	-20.3	83.1	5.3	3.4	-24.1	85.5	-27.2	87.5	-30.2	89.5	-34.1	91.9	(4)
(5)			WB	-20.7	68.2	5.2	2.4	-24.4	70.0	-27.5	71.4	-30.4	72.8	-34.2	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	17.6	19.5	26.6	36.7	46.5	54.1	59.3	56.3	48.4	37.5	28.3	20.5
	DBStd	16.90	13.02	12.67	9.70	6.22	6.91	5.73	5.21	5.34	5.74	6.49	8.81	12.60
	HDD50	5219	1005	855	725	400	151	22	0	9	96	389	651	916
	HDD65	9980	1470	1275	1190	850	573	329	190	273	497	852	1100	1381
	CDD50	735	0	0	0	0	44	146	289	205	49	2	0	0
	CDD65	20	0	0	0	0	0	2	14	4	0	0	0	0
	CDH74	401	0	0	0	0	21	85	233	61	1	0	0	0
	CDH80	51	0	0	0	0	0	6	42	3	0	0	0	0
Wind	WSAvg	3.9	2.9	3.3	4.3	4.3	5.0	5.1	4.2	3.9	3.8	3.5	3.3	3.0
Precipitation	PrecAvg	23.9	1.5	1.0	1.0	1.3	1.8	2.6	3.7	3.3	2.0	2.2	1.8	1.6
	PrecMax	34.4	2.7	2.1	1.9	3.3	3.5	4.8	9.2	5.9	4.4	6.4	3.9	3.0
	PrecMin	18.1	0.2	0.0	0.4	0.3	0.5	0.7	0.7	1.1	0.6	0.6	0.6	0.2
	PrecStd	3.9	0.6	0.5	0.4	0.7	0.8	1.0	2.0	1.4	1.0	1.3	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	44.8	49.6	65.2	77.0	80.7	84.8	79.3	69.6	57.8	50.7	44.2
		MCWB	38.6	39.6	41.4	49.9	58.5	61.3	65.7	64.8	59.4	51.5	46.8	40.7
	2%	DB	39.5	41.9	45.6	58.0	71.6	76.5	79.9	75.2	65.0	53.0	45.5	40.8
		MCWB	36.5	37.7	39.2	46.9	56.2	59.3	63.2	62.7	57.1	49.5	43.1	37.8
	5%	DB	36.3	38.9	42.7	53.3	66.4	71.2	76.0	71.5	61.8	50.5	42.0	38.4
		MCWB	34.0	35.7	37.3	44.1	53.3	57.5	62.5	61.2	55.3	47.5	40.1	35.9
	10%	DB	33.8	35.7	40.2	49.1	61.2	67.4	72.2	67.9	58.7	48.1	39.6	35.7
		MCWB	32.2	33.2	35.6	41.4	51.2	56.0	60.6	59.6	53.5	45.1	37.9	34.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	40.1	42.3	51.4	60.8	64.4	68.0	67.5	61.1	52.9	47.2	41.2
		MCDB	41.7	43.6	47.8	62.3	70.7	76.5	79.7	75.0	67.1	56.1	50.4	43.9
	2%	WB	36.6	38.1	40.0	47.3	57.8	61.1	65.6	64.7	58.3	50.1	43.4	38.2
		MCDB	39.0	41.4	44.9	57.2	69.0	70.8	75.9	71.7	63.4	52.2	45.3	40.4
	5%	WB	34.3	35.8	37.9	44.8	54.8	59.0	63.8	62.3	56.2	48.0	40.5	36.0
		MCDB	35.9	38.5	42.2	52.0	64.7	69.2	73.1	69.5	60.7	50.4	41.7	37.8
	10%	WB	32.3	33.3	35.7	42.1	51.7	57.0	61.8	60.3	54.2	45.3	38.0	34.0
		MCDB	33.7	35.4	39.6	48.2	60.3	65.9	70.2	66.4	57.9	47.8	39.4	35.5
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.2	17.2	18.4	20.2	19.2	19.1	18.6	16.3	12.0	9.4	12.1
		MCDBR	13.7	14.3	17.5	27.7	31.0	27.7	27.1	24.9	20.9	15.4	11.7	13.0
		MCWBR	12.4	11.7	12.1	17.2	16.9	13.7	13.2	13.9	13.7	11.9	10.3	11.3
	5% WB	MCDBR	13.2	14.3	16.9	25.4	27.6	24.0	23.4	21.4	18.6	14.3	11.4	12.4
		MCWBR	12.1	11.9	12.2	16.4	15.9	13.0	12.4	12.7	12.8	11.7	10.2	10.9
Clear-Sky Solar Irradiance	taub	0.220	0.281	0.307	0.343	0.347	0.343	0.357	0.348	0.317	0.301	N/A	0.348	
	taud	2.053	2.222	2.302	2.372	2.443	2.482	2.457	2.478	2.553	2.457	N/A	2.683	
	Ebn at Noon	132	209	252	266	276	279	271	263	252	206	N/A	78	
	Edh at Noon	10	19	26	30	31	30	30	27	21	16	N/A	6	
All-Sky Solar Radiation	RadAvg	57	256	709	1209	1499	1716	1571	1188	725	313	85	28	
	RadStd	6	38	76	99	160	120	172	103	82	51	16	2	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-223	N/A	N/A	N/A	

Nomenclature: See separate page